

## SANDERSON FIELD, WA, USA

WMO: 727925

Lat: 47.238N

Lon: 123.141W

Elev: 83

StdP: 100.34

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 94227

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | -5.5       | -3.4 | -10.4                         | 1.6 | -2.5 | -7.2 | 2.1 | -0.7 | 11.0                  | 8.8  | 9.3 | 8.5  | 0.4                   | 40   | 0.570 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 13.4                      | 31.3            | 18.7 | 28.8 | 18.0 | 26.6 | 17.2 | 19.7                | 29.3 | 18.7 | 27.6 | 17.7 | 25.3 | 3.2                     | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.2                            | 11.6 | 21.0 | 15.3 | 11.0 | 20.5 | 14.6 | 10.5 | 19.8 | 56.0          | 29.3 | 53.0 | 27.6 | 49.9 | 25.3 | 26.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| 9.2               | 8.2  | 7.3 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
|                   |      |     | -9.0                       | 35.5 | 3.0                | 2.1 | -11.2                                              | 37.1 | -12.9      | 38.3 | -14.6      | 39.5 | -16.8      | 41.1 |  |  |
|                   |      |     | -9.6                       | 21.8 | 2.9                | 1.8 | -11.7                                              | 23.1 | -13.4      | 24.2 | -15.0      | 25.2 | -17.1      | 26.5 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.4        | 4.3   | 4.8   | 6.6   | 8.8   | 12.4  | 15.1  | 18.0  | 18.2  | 15.3  | 10.4  | 6.4   | 3.9   |
|                                                                        | DBStd       | 5.87        | 3.23  | 2.80  | 2.64  | 2.85  | 3.06  | 2.91  | 2.60  | 2.38  | 2.58  | 2.74  | 3.44  | 3.23  |
|                                                                        | HDD10.0     | 838         | 178   | 147   | 110   | 55    | 11    | 0     | 0     | 0     | 1     | 29    | 117   | 191   |
|                                                                        | HDD18.3     | 2976        | 435   | 379   | 364   | 285   | 186   | 105   | 38    | 32    | 96    | 247   | 359   | 448   |
|                                                                        | CDD10.0     | 972         | 1     | 1     | 4     | 20    | 84    | 153   | 247   | 254   | 159   | 40    | 7     | 1     |
|                                                                        | CDD18.3     | 69          | 0     | 0     | 0     | 0     | 1     | 8     | 27    | 28    | 5     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 1362        | 0     | 0     | 1     | 14    | 74    | 183   | 461   | 459   | 166   | 2     | 1     | 0     |
|                                                                        | CDH26.7     | 482         | 0     | 0     | 0     | 2     | 18    | 63    | 176   | 174   | 49    | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 2.6         | 2.0   | 2.2   | 2.6   | 2.9   | 3.2   | 3.5   | 3.5   | 3.2   | 2.4   | 2.1   | 2.0   | 1.9   |
| (15) Precipitation                                                     | PrecAvg     | 1630        | 263   | 174   | 184   | 115   | 71    | 47    | 19    | 35    | 58    | 165   | 269   | 262   |
|                                                                        | PrecMax     | 2174        | 553   | 496   | 441   | 251   | 142   | 89    | 68    | 103   | 228   | 408   | 644   | 466   |
|                                                                        | PrecMin     | 1224        | 119   | 0     | 35    | 37    | 6     | 7     | 3     | 0     | 0     | 5     | 93    | 81    |
|                                                                        | PrecStd     | 287         | 94    | 91    | 83    | 54    | 40    | 24    | 17    | 30    | 56    | 97    | 122   | 102   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 12.9  | 14.7  | 20.1  | 25.4  | 29.1  | 32.3  | 35.2  | 34.2  | 31.2  | 22.7  | 16.0  | 13.5  |
|                                                                        |             | MCWB        | 11.3  | 8.3   | 11.9  | 14.3  | 16.6  | 19.3  | 20.1  | 19.5  | 17.7  | 14.1  | 13.8  | 11.9  |
|                                                                        | 2%          | DB          | 11.3  | 12.4  | 16.2  | 20.9  | 25.4  | 28.3  | 31.1  | 31.1  | 27.7  | 19.2  | 13.9  | 11.2  |
|                                                                        |             | MCWB        | 10.0  | 8.4   | 10.3  | 12.6  | 15.6  | 17.8  | 18.8  | 19.0  | 17.1  | 13.0  | 11.9  | 10.0  |
|                                                                        | 5%          | DB          | 10.2  | 10.9  | 13.4  | 17.4  | 22.0  | 25.0  | 28.1  | 28.3  | 24.8  | 17.1  | 12.6  | 9.8   |
|                                                                        |             | MCWB        | 8.9   | 8.2   | 9.0   | 10.7  | 14.3  | 16.4  | 18.3  | 18.3  | 16.3  | 12.9  | 11.1  | 8.6   |
|                                                                        | 10%         | DB          | 9.1   | 9.5   | 11.5  | 14.7  | 19.0  | 22.0  | 25.6  | 25.6  | 22.1  | 15.3  | 11.4  | 8.5   |
|                                                                        |             | MCWB        | 7.9   | 7.6   | 8.4   | 9.6   | 12.7  | 15.0  | 17.3  | 17.4  | 15.3  | 12.6  | 10.1  | 7.4   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 12.0  | 11.1  | 12.9  | 15.4  | 17.7  | 20.1  | 21.5  | 20.7  | 19.5  | 16.2  | 14.1  | 12.2  |
|                                                                        |             | MCDB        | 12.6  | 12.4  | 17.3  | 23.5  | 26.5  | 30.6  | 32.8  | 31.2  | 27.5  | 18.9  | 15.7  | 13.2  |
|                                                                        | 2%          | WB          | 10.5  | 9.9   | 11.1  | 13.0  | 16.0  | 18.3  | 19.6  | 19.6  | 17.9  | 14.8  | 12.5  | 10.2  |
|                                                                        |             | MCDB        | 11.0  | 11.0  | 14.5  | 19.8  | 23.8  | 27.1  | 29.2  | 29.4  | 24.9  | 17.1  | 13.5  | 11.0  |
|                                                                        | 5%          | WB          | 9.1   | 8.7   | 10.0  | 11.5  | 14.6  | 16.6  | 18.6  | 18.6  | 16.9  | 13.8  | 11.5  | 8.8   |
|                                                                        |             | MCDB        | 9.8   | 10.1  | 12.5  | 15.8  | 21.1  | 23.8  | 27.6  | 27.3  | 22.7  | 15.9  | 12.4  | 9.6   |
|                                                                        | 10%         | WB          | 8.0   | 7.8   | 9.0   | 10.3  | 13.4  | 15.3  | 17.5  | 17.7  | 16.0  | 12.8  | 10.3  | 7.7   |
|                                                                        |             | MCDB        | 8.8   | 9.2   | 11.2  | 13.7  | 18.3  | 21.2  | 25.0  | 25.0  | 20.9  | 14.8  | 11.2  | 8.4   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.3   | 8.1   | 9.2   | 10.6  | 11.3  | 11.5  | 13.2  | 13.4  | 12.8  | 9.8   | 7.5   | 5.9   |
|                                                                        |             | MCDBR       | 7.1   | 11.0  | 13.8  | 17.2  | 18.3  | 18.6  | 19.7  | 20.2  | 19.8  | 13.7  | 8.4   | 6.2   |
|                                                                        | 5% WB       | MCWBR       | 5.7   | 7.4   | 8.4   | 9.4   | 9.5   | 8.8   | 8.9   | 9.1   | 10.0  | 8.6   | 6.2   | 5.2   |
|                                                                        |             | MCWBR       | 5.8   | 7.8   | 11.0  | 13.6  | 15.6  | 16.2  | 18.0  | 17.4  | 15.7  | 10.2  | 6.5   | 5.6   |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.331 | 0.321 | 0.336 | 0.367 | 0.393 | 0.376 | 0.361 | 0.363 | 0.348 | 0.338 | 0.335 | 0.331 |
|                                                                        |             | taud        | 2.446 | 2.526 | 2.497 | 2.378 | 2.322 | 2.399 | 2.463 | 2.460 | 2.504 | 2.529 | 2.457 | 2.397 |
|                                                                        | Ebn at Noon | Ebn at Noon | 749   | 853   | 890   | 887   | 872   | 888   | 897   | 881   | 865   | 820   | 736   | 694   |
|                                                                        |             | Edh at Noon | 68    | 76    | 91    | 112   | 123   | 115   | 106   | 102   | 89    | 75    | 67    | 64    |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 1.02        | 1.87  | 2.79  | 4.09  | 5.18  | 5.66  | 6.33  | 5.41  | 3.97  | 2.25  | 1.19  | 0.84  | (44)  |
|                                                                        | RadStd      | 0.15        | 0.26  | 0.34  | 0.39  | 0.43  | 0.58  | 0.48  | 0.38  | 0.37  | 0.25  | 0.16  | 0.12  | (45)  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | -0.80   | -2.15  | N/A     | N/A   | N/A   | N/A         | N/A     | +143    | +31     |
| Regional (58 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page